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Systems and methods for robotic arm end effector for tailored through thickness reinforcement

Abstract

An end effector for needling a fibrous preform includes a body (or head), a plurality of articulating presser feet mounted to the body and moveable with respect to the body, and a plurality of articulating needles mounted to the body and moveable with respect to the body. The needles may be configured to move between an extended position and a retracted position. The presser feet may be configured to move between an extended position and a retracted position. The end effector presser feet may conform to complex geometries of a fibrous preform. The end effector may control needling depth, needling density, applied tension and/or pressure on the fibrous preform, and needle angle.

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Background/Summary

FIELD

(1) The present disclosure relates generally to the use of robotic arm end effectors during the preparing of materials during the process of laying up and manufacture and, more particularly, to through-thickness reinforcement of complex contour materials.

BACKGROUND

(2) Composites with through-thickness reinforcement generally provide higher interlaminar properties but lower in-plane properties compared to 2D lay-ups. Composite structures for aerospace applications sometimes have complex contours to meet aerodynamic and thermal requirements. Additionally, it is desirable for composite structures to be designed to withstand in-plane and interlaminar stresses. These stresses can vary across the structure.

SUMMARY

(3) An end effector for through thickness reinforcement of a fibrous preform is disclosed, comprising a body, a first presser foot mounted to the body and moveable with respect to the body, and a first needle mounted to the body and moveable with respect to the body. The first needle is configured to move between a first extended position and a first retracted position.

(4) In various embodiments, the first presser foot is configured to move between a second extended position and a second retracted position.

- (5) In various embodiments, the end effector further comprises a second presser foot, wherein the first presser foot is configured to articulate independent of the second presser foot.
- (6) In various embodiments, the first presser foot is moveable with respect to the first needle.
- (7) In various embodiments, the end effector further comprises a second needle and a second presser foot. The first needle extends through the first presser foot and the second needle extends through the second presser foot.
- (8) In various embodiments, the first presser foot comprises a pivoting foot section pivotally mounted to a spherical joint.
- (9) In various embodiments, the end effector further comprises a flexible material provided at an end of the pivoting foot section, wherein the flexible material extends between adjacent pivoting foot sections.
- (10) In various embodiments, the end effector further comprises a spring member configured to bias the first presser foot.
- (11) In various embodiments, the end effector further comprises an actuator for moving the first presser foot between the second extended position and the second retracted position.
- (12) In various embodiments, the end effector further comprises an actuator for moving the first needle between the first extended position and the first retracted position.
- (13) In various embodiments, the end effector further comprises a preform tensioner configured to contact the fibrous preform to apply tension to the fibrous preform while the first needle moves between the first extended position to the first retracted position.
- (14) A method for performing a process of through-thickness reinforcement in a fibrous preform is disclosed, in accordance with various embodiments. The method comprises moving an end effector with respect to the fibrous preform, moving the end effector toward the fibrous preform until a first presser foot of the end effector contacts the fibrous preform, moving a first needle of the end effector from a first retracted position to a first extended position, penetrating the fibrous preform with the first needle in response to the first needle moving from the first retracted position to the first extended position, and moving at least a portion of a fiber from a first layer of the fibrous preform into a second layer of the fibrous preform in response to the first needle penetrating the fibrous preform.
- (15) In various embodiments, the method further comprises receiving, by a control unit, a feedback signal from a sensor indicating a pressure exerted by the first presser foot onto the fibrous preform.
- (16) In various embodiments, the method further comprises orienting the first needle at a non-orthogonal angle with respect to the fibrous preform. The fibrous preform may have a contoured surface (i.e., is not flat) at the location of the first needle.
- (17) In various embodiments, the method further comprises moving a second presser foot of the end effector from a second retracted position to a second extended position.
- (18) In various embodiments, the method further comprises receiving, by a control unit, a feedback signal from a sensor indicating a pressure exerted by the first needle onto the fibrous preform.
- (19) A method for performing a process of through-thickness reinforcement in a fibrous preform is disclosed, in accordance with various embodiments. The method comprises moving an end effector with respect to the fibrous preform, moving the end effector toward the fibrous preform until a first presser foot of the end effector contacts the fibrous preform, receiving a first pressure signal from a first sensor indicating a first pressure applied by the first presser foot onto the fibrous preform, moving a second presser foot of the end effector from a first retracted position to a first extended position, receiving a second pressure signal from a second sensor indicating a second pressure applied by the second presser foot onto the fibrous preform, moving a first needle of the end effector from a second retracted position to a second extended position, moving a second needle of the end effector from a third retracted position to a third extended position, penetrating the fibrous preform with the first needle in response to the first needle moving from the second retracted position to the second extended position, penetrating the fibrous preform with the second needle in

response to the second needle moving from the third retracted position to the third extended position, moving at least a portion of a first fiber from a first layer of the fibrous preform into a second layer of the fibrous preform in response to the first needle penetrating the fibrous preform, moving at least a portion of a second fiber from the first layer of the fibrous preform into the second layer of the fibrous preform in response to the second needle penetrating the fibrous preform.

(20) In various embodiments, the method further comprises moving the end effector toward the fibrous preform until the first pressure signal indicates the first pressure applied by the first presser foot is equal to or greater than a predetermined pressure threshold.

(21) In various embodiments, the method further comprises continuing to extend the second presser foot until the second pressure signal indicates the second pressure applied by the second presser foot is equal to or greater than the predetermined pressure threshold.

(22) In various embodiments, the method further comprises pivoting the first presser foot in response to the first presser foot contacting the fibrous preform.

(23) In various embodiments, the method further comprises penetrating the fibrous preform at a first depth in response to moving the first needle to the second extended position, and penetrating the fibrous preform at a second depth in response to moving the second needle of the end effector to the third extended position, wherein the first depth is different from the second depth.

(24) In various embodiments, the needle may be configured with one or more barbs along the length of the needle, wherein each barb is designed to entrain or capture one or more fibrous filaments within a ply or layer of the fibrous preform. In various embodiments, as the needle penetrates the fibrous preform, at least a portion of the entrained fibrous filaments in the barbs are transported along the direction of the penetrating needle to provide through-thickness reinforcement. In various embodiments, the needle may be alternatively or additionally configured to be a stitching or a tufting needle with an eye to transport fibrous filament along the direction of the penetration.

(25) The foregoing features and elements may be combined in any combination, without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims.

(2) FIG. 1 is a perspective illustration of an exemplary robotic arm comprising an end effector during a through thickness reinforcement process of a fibrous preform, in accordance with various embodiments;

(3) FIG. 2 is an illustration of a face of the end effector of FIG. 1, in accordance with various embodiments;

(4) FIG. 3 is a schematic sectional illustration of the end effector during a through thickness reinforcement process, in accordance with various embodiments;

(5) FIG. 4A, FIG. 4B, and FIG. 4C are schematic sectional illustrations of the end effector during a through thickness reinforcement process with the end effector nominally normal to a flat portion of

the fibrous preform, in accordance with various embodiments;

(6) FIG. 5A, FIG. 5B, and FIG. 5C are schematic sectional illustrations of the end effector during a through thickness reinforcement process with the end effector at an angle with respect to a flat portion of the fibrous preform, in accordance with various embodiments;

(7) FIG. 6A, FIG. 6B, and FIG. 6C are schematic sectional illustrations of the end effector during a through thickness reinforcement process with the end effector nominally normal to a rounded portion of the fibrous preform, in accordance with various embodiments;

(8) FIG. 7A, FIG. 7B, and FIG. 7C are schematic sectional illustrations of the end effector during a through thickness reinforcement process with the end effector at an angle with respect to a rounded portion of the fibrous preform, in accordance with various embodiments;

(9) FIG. 8 is a schematic sectional illustration of an end effector having presser feet that include pivoting foot sections, in accordance with various embodiments;

(10) FIG. 9 is a schematic sectional illustration of an end effector having presser feet that include pivoting foot sections pivotally mounted to spherical joints, in accordance with various embodiments;

(11) FIG. 10 is a schematic sectional illustration of an end effector having a flexible material coupled to the ends of the presser feet, in accordance with various embodiments;

(12) FIG. 11 is a schematic sectional illustration of an end effector having an array of retractable presser feet disposed adjacent and alternating with an array of retractable needles, in accordance with various embodiments;

(13) FIG. 12 is a schematic sectional illustration of an end effector having an actively controlled array of retractable presser feet, in accordance with various embodiments;

(14) FIG. 13 is a schematic sectional illustration of an end effector having a passively controlled array of retractable presser feet, in accordance with various embodiments; and

(15) FIG. 14 and FIG. 15 are a schematic sectional illustrations of end effectors having preform tensioners, in accordance with various embodiments.

DETAILED DESCRIPTION

(16) The following detailed description of various embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that changes may be made without departing from the scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. It should also be understood that unless specifically stated otherwise, references to “a,” “an” or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. Further, all ranges may include upper and lower values and all ranges and ratio limits disclosed herein may be combined.

(17) As used herein, the term “through thickness reinforcement” includes “needling,” “stitching,” and/or “tufting,” in accordance with various embodiments.

(18) As used herein, the term “needling” includes “stitching” and/or “tufting,” in accordance with various embodiments.

(19) With reference to FIG. 1, a robotic arm **100** comprising an end effector **102** is illustrated during a through thickness reinforcement process of a fibrous preform **110**, in accordance with various embodiments. Fibrous preform **110** may be placed over a tool **115** and formed to the geometry of the tool **115**. In this manner, fibrous preform **110** may be shaped into a complex

contour preform (e.g., bent about one or more axes). Robotic arm **100** may be configured to move the end effector **102** with respect to the fibrous preform **110** in a controlled manner to perform a through thickness reinforcement process, i.e. Z-needling. Z-needling refers to a process comprising penetrating a composite material (e.g., fibrous preform **110**) with needles and moving (e.g., by pulling or pushing) fibers from the in-plane direction and forcing them into the Z direction, where the “Z direction” as used herein refers to a direction perpendicular to the in-plane direction. For preforms having curved surfaces, the “Z-direction” refers to the direction normal to a (local) surface of the fibrous preform at the point where the preform is being needled (i.e., a direction normal to the tangent plane to the surface at the point of needling). In general, the through thickness reinforcement process has the effect of interlocking individual fabric layers together. Thus, after through thickness reinforcement, the fibrous material has fibers extending in three different directions (i.e., in the X and Y directions in the plane of the fibrous layers and the Z direction perpendicular to the fibrous layers). It should be appreciated that due to the complex contours of the fibrous preform **110**, the X, Y, and Z directions vary depending on the particular location of the fibrous preform **110**.

(20) With reference to FIG. 2, a view of the face of end effector **102** is illustrated, in accordance with various embodiments. End effector **102** may comprise a head or body **104**, a plurality of presser feet **106** (e.g., first presser foot **106a**, second presser foot **106b**, etc.), and a plurality of needles **108** (e.g., first needle **108a**, second needle **108b**, etc.). Body **104** may be made from a metal material, a composite material, or a plastic material. Body **104** may house various components for actuating presser feet **106** and/or needles **108**. Presser feet **106** may be moveable with respect to body **104**. Presser feet **106** may be moveable with respect to needles **108**. Needles **108** may be moveable with respect to body **104**. Needles **108** may be moveable with respect to presser feet **106**. A face **105** of the body may be flat or planar. Face **105** may face the fibrous preform during the through thickness reinforcement process. In various embodiments, presser feet **106** and needles **108** extend from face **105**.

(21) In various embodiments, each presser foot **106** comprises a cylindrical body with a hemispherical-shaped end; though other presser foot **106** shapes and/or designs are contemplated herein. Each presser foot **106** may be made from a metal material, a rubber material, or a plastic material. The material of presser foot **106** may be selected based on the desired pressure exerted onto fibrous preform **110**, among other factors.

(22) In various embodiments, each needle **108** may be made from a metal material or a polymer material or a ceramic material or combinations thereof. For example, the needle may be a metal material with a polymer coating or it may be a metal material with a ceramic coating. The material and coating system of needle **108** may be selected depending on the material of fibrous preform **110**, among other factors.

(23) With reference to FIG. 3, end effector **102** is illustrated during a through thickness reinforcement process of fibrous preform **110**. Fibrous preform **110** may comprise a first layer **112** and a second layer **114**. First layer **112** may be a top layer. Each layer of material may share a common (e.g., the same) construction and/or material makeup. Each layer of material, for example, may be formed by a sheet/layer of fibrous material; e.g., woven carbon fiber, woven oxidized polyacrylonitrile (PAN) fibers, non-crimp fabric, etc. One or more or all of the layers of material may each be impregnated with a polymer matrix; e.g., thermoset material or thermoplastic material. One or more or all of the layers of material may alternatively each be unimpregnated (e.g., only include the fibrous material) where, for example, the preform material is impregnated subsequent to formation of the composite structure. The method of the present disclosure, however, is not limited to such exemplary layer materials.

(24) End effector **102** may be moved (e.g., via robotic arm **100**) with respect to fibrous preform **110**. With the end effector **102** in the desired position, one or more needles may be actuated to penetrate fibrous preform **110**, thereby moving one or more fibers **113** from first layer **112** into

second layer **114** and interlocking first layer **112** with second layer **114**. For example, needle **108a** is illustrated in FIG. 3 moving from a non-penetrating position to a penetrating position (also referred to herein as a retracted position (see needle **108b**) and an extended position (see needle **108a**), respectively). In this regard, needles **108** may be referred to herein as articulating needles. Needle **108b** may be similarly operated. Needles **108a** and **108b** can be controlled individually or in groups by programmable robotic system (e.g., see control unit **250** of FIG. 5B and FIG. 5C) to puncture the plies of fibrous preform **110** to a desired depth and/or a desired needling density (e.g., various needles **108** may be commanded not to penetrate the fibrous preform to vary a needling density (i.e., number of needles per unit area)). The needles **108a** and **108b** may be configured to puncture the fibers in the top ply or a sacrificial ply layer into the adjacent plies at the desired angle and depth. The end effector **102** may be rotated to appropriate angles to needle plies at different desired angles (e.g., see FIG. 4A through FIG. 7C, FIG. 14, and FIG. 15).

(25) FIG. 3 illustrates presser foot **106a** in a retracted position and presser foot **106b** in an extended position, whereby the presser feet **106**. In this regard, presser feet **106** may be referred to herein as articulating presser feet.

(26) With reference to FIG. 4A, FIG. 4B, and FIG. 4C, an end effector **202** is illustrated during a through thickness reinforcement process of fibrous preform **210**, in accordance with various embodiments. End effector **202** may be similar to end effector **102** (see FIG. 1) in various embodiments. FIG. 4A illustrates end effector **202** moved with respect to fibrous preform **210**. For example, end effector **202** may be oriented at a desired angle with respect to fibrous preform **210**. End effector **202** may be moved toward fibrous preform **210** until one or more presser feet **206** contact fibrous preform **210**. For example, presser feet **206** may exert a desired pressure on fibrous preform **210** to secure fibrous preform **210** during the through thickness reinforcement process.

(27) Each needle **208** may be actuated independently so as to independently control a penetrating depth of each needle **208**. For example, a first needle **208a** may be actuated to penetrate a first predetermined number of layers of fibrous preform **210**, such as three layers for example, a second needle **208b** may be actuated to penetrate a second predetermined number of layers of fibrous preform **210**, such as six layers for example, and a third needle **208c** may be actuated to penetrate a third predetermined number of layers of fibrous preform **210**, such as two layers for example. Although penetration of three, six, and two layers is provided as an example, any number of layers of fibrous preform **210** may be penetrated by needles **208** as desired depending on the desired amount of interlocking of adjacent layers. In this regard, first needle **208a** may be actuated to penetrate a first number of layers of the fibrous preform and the second needle **208b** may be simultaneously, or nearly simultaneously, actuated to penetrate a second number of layers of the fibrous preform **210**, wherein the first number of layers is different from the second number of layers. Stated differently, first needle **208a** may penetrate the fibrous preform **210** at a first depth and second needle **208b** may penetrate the fibrous preform **210** at a second depth, different from the first depth. For example, one or more layers be omitted from the through thickness reinforcement process to allow for said layer(s) to move with respect to one another during the forming process to avoid wrinkling. In various embodiments, one or more layers may be omitted from the through thickness reinforcement process depending on the desired interlaminar preform mechanical properties, through thickness thermal conductivity, and/or other fiber dominated properties. Moreover, by independently controlling each needle, the frequency of needling may be varied as desired. Having discussed the independent control of the needles **208**, the presser feet **206** may be similarly independently controlled.

(28) With reference to FIG. 5A, FIG. 5B, and FIG. 5C, end effector **202** is illustrated during a through thickness reinforcement process of fibrous preform **210**, in accordance with various embodiments. FIG. 5A illustrates end effector **202** moved with respect to fibrous preform **210**. FIG. 5B illustrates end effector **202** oriented at a desired angle α with respect to fibrous preform **210**. In various embodiments, angle α is an angle of the centerline of a needle **208** with respect to the

fibrous preform. Angle α may be between twenty degrees and ninety degrees in accordance with various embodiments, between thirty degrees and eighty degrees in accordance with various embodiments, or between twenty five degrees and eighty degrees in accordance with various embodiments. End effector **202** may be moved toward fibrous preform **210** until one or more presser feet **206** contact fibrous preform **210**. For example, presser feet **206** may exert a desired pressure on fibrous preform **210** to secure fibrous preform **210** during the through thickness reinforcement process.

(29) In various embodiments, the end effector **202** further includes a control unit **250**, which includes one or more controllers (e.g., processors) and one or more tangible, non-transitory memories capable of implementing digital or programmatic logic. In various embodiments, for example, the one or more controllers are one or more of a general purpose processor, digital signal processor (DSP), application specific integrated circuit (ASIC), field programmable gate array (FPGA), or other programmable logic device, discrete gate, transistor logic, or discrete hardware components, or any various combinations thereof or the like. In various embodiments, the control unit **250** controls, at least various parts of, and operation of various components of, the end effector **202**. For example, the control unit **250** may control a position of end effector **202** with respect to fibrous preform **210**, the position of presser feet **206** (e.g., via actuators **222a**, **222b**, **222c**), the position of needles **208** (e.g., via actuators **224a**, **224b**, **224c**), and may receive feedback from sensors **220a**, **220b**, **220c**.

(30) In various embodiments, end effector **202** further includes a touch or pressure sensor for each presser foot **206**. For example, first presser foot **206a** may comprise a first sensor **220a**, second presser foot **206b** may comprise a second sensor **220b**, and third presser foot **206c** may comprise a third sensor **220c**. In various embodiments, the robotic arm moves the end effector **202** toward the fibrous preform **210** until at least one sensor (e.g., sensor **220a**) detects that presser foot **206a** has contacted fibrous preform **210**. In various embodiments, control unit **250** detects (via sensor **220a**) that presser foot **206a** has contacted fibrous preform **210** and is exerting a predetermined force or pressure on the fibrous preform **210**, which may ensure that presser foot **206a** secures the fibrous preform **210** during the through thickness reinforcement process. For example, the robotic arm may monitor the pressure feedback signal from sensor **220a** and continue to press the presser foot **206a** into fibrous preform **210** until the pressure feedback signal indicates that presser foot **206a** is exerting a pressure greater than or equal to a predetermined pressure threshold.

(31) In various embodiments, once at least one presser foot **206a** contacts the fibrous preform **210**, end effector **202** may be actuated to extend presser foot **206b** and presser foot **206c** until control unit **250** detects (via sensors **220b** and **220c**) that presser feet **206b** and **206c**, respectively, have contacted fibrous preform **210** (and in various embodiments detect that presser feet **206b** and **206c** are exerting a predetermined force or pressure on the fibrous preform **210**). In this regard, each sensor **220a**, **220b**, and **220c**—schematically illustrated in FIG. 5B—may comprise a touch sensors and/or a pressure sensor. In various embodiments, sensors **220a**, **220b**, **220c** may comprise multi-axial sensors to measure force in different directions.

(32) In various embodiments, end effector **202** further includes an actuator for each presser foot **206**. For example, each presser feet **206a**, **206b**, and **206c** may comprise actuators **222a**, **222b**, and **222c**, respectively, for extending and/or retracting the respective presser foot **206a**, **206b**, and **206c**. Actuators **222a**, **222b**, and **222c**—schematically illustrated in FIG. 5B—may be linear actuators or rotary actuators. Actuators **222a**, **222b**, and **222c** may be pneumatic actuators, electric actuators, or hydraulic actuators.

(33) In various embodiments, control unit **250** receives feedback from sensors **220a**, **220b**, **220c** for closed-loop control of the position of presser feet **206**. For example, presser foot **206a** may be extended until a pressure feedback signal received from the associated sensor **220a** indicates that the presser foot **206a** is exerting a predetermined pressure on the fibrous preform **210**.

(34) FIG. 5C illustrates the needles **208** in extended positions during the through thickness

reinforcement process. In various embodiments, end effector **202** further includes an actuator for each needle **208**. In various embodiments, each actuator may actuate a single needle or a zone of needles (e.g., a row of needles or a column of needles in accordance with various embodiments). Needles **208** may be actuated independent of the position of the presser feet **206**. In the illustrated embodiments, needles **208a**, **208b**, and **208c** comprise actuators **224a**, **224b**, and **224c**, respectively, for extending and/or retracting the respective needle **208a**, **208b**, and **208c**.

(35) In various embodiments, control unit **250** may monitor pressure exerted by the needles **208** to alert for needle breaking and/or proper loading of fibers of the fibrous preform **210**.

(36) With reference to FIG. **6A**, FIG. **6B**, and FIG. **6C**, end effector **202** may be particularly useful for portions of fibrous preform **210** which are rounded or have complex contours due to the ability of the presser feet **206** to conform to a curved surface of the fibrous preform **210**. In the illustrated embodiment, and with particular focus on FIG. **6B**, end effector **202** may be oriented nominally normal to the surface **211** of the fibrous preform **210**. In various embodiments, a centrally located presser foot **206b** may first contact the surface **211** and the surrounding presser feet **206a** and **206c** may be actuated to extend to contact the surface **211** to provide uniform pressure on the fibrous preform during the through thickness reinforcement process (see FIG. **6C**); though in various embodiments a non-centrally located presser foot may first contact the fibrous preform **210** and centrally located presser feet may be actuated to contact the fibrous preform **210**.

(37) With particular focus on FIG. **6C**, needles **208** and/or presser feet **206** may be configured to articulate (e.g., pivot) to maintain needles **208** and/or presser feet **206** nominally perpendicular to the surface **211** of the fibrous preform **210**. For example, needles **208** and/or presser feet **206** may be mounted to a pivot, such as a ball joint or the like, to accommodate rotation of the needles **208** and/or presser feet **206** with respect to body **204** of end effector **202**.

(38) With reference to FIG. **7A**, FIG. **7B**, and FIG. **7C**, a through thickness reinforcement process using end effector **202** is illustrated at a curved portion of fibrous preform **210** similar to FIG. **6A** through FIG. **6C**, except that end effector **202** is oriented at an angle with respect to the surface **211** of the fibrous preform **210**. In this regard, needles **208** may penetrate the plies of the fibrous preform **210** at a non-orthogonal angle during the through thickness reinforcement process.

(39) With reference to FIG. **8**, an end effector **302** is illustrated, in accordance with various embodiments. End effector **302** may be similar to end effector **102** (see FIG. **1**) in various embodiments. End effector **302** comprises a plurality of needles **308** and a plurality of pivoting presser feet **306**, in accordance with various embodiments. Each pivoting presser foot **306** may comprise a support leg **326** mounted to body **304** and a pivoting foot section **328** pivotally mounted to support leg **326** via a pivot **330**. In this manner, pivoting foot section **328** may pivot to apply uniform pressure on the fibrous preform during the through thickness reinforcement process. In various embodiments, needle **308** extends through a portion of pivoting foot section **328**.

(40) With reference to FIG. **9**, an end effector **402** is illustrated, in accordance with various embodiments. End effector **402** may be similar to end effector **102** (see FIG. **1**) in various embodiments. End effector **402** comprises a plurality of needles **408** and a plurality of pivoting presser feet **406**, in accordance with various embodiments. Each pivoting presser foot **406** may comprise a spherical joint **426** mounted to body **404** and a pivoting foot section **428** pivotally mounted to spherical joint **426**. In this manner, pivoting foot section **428** may pivot to apply uniform pressure on the fibrous preform during the through thickness reinforcement process. Spherical joint **426** may accommodate multi-axial rotation of pivoting foot section **428**.

(41) Needle **408** may extend through spherical joint **426**. Needle **408** may extend through pivoting foot section **428**. In this regard, pivoting foot section **428** may comprise an aperture **432** extending therethrough. Aperture **432** may be tapered to accommodate needle **408** (i.e., provide clearance for needle **408**) as pivoting foot section **428** pivots about spherical joint **426**. A spring member **434** may bias pivoting foot section **428** to a centered position (e.g., wherein the central axis of pivoting foot section **428** is generally coaxial with the central axis of needle **408**). Spring member **434** may

be tailored to provide a desired amount of pressure on the fibrous preform during the through thickness reinforcement process. Needle **408** and spherical joint **426** may extend at least partially through spring member **434**. In various embodiments, spring member **434** is a coil spring. It is further contemplated that spring member **434** may be replaced with a solid link that is mechanically or electrically controlled to articulate pivoting foot section **428**.

(42) With reference to FIG. **10**, an end effector **502** is illustrated, in accordance with various embodiments. End effector **502** may be similar to end effector **102** (see FIG. **1**) in various embodiments. End effector **502** may be similar to end effector **402** except that end effector **502** further includes a conformable, flexible material **536** extending between and to adjacent pivoting foot sections **428** to create a contiguous or discontinuous surface for uniform compression of the fibrous preform during the through thickness reinforcement process. In this manner, flexible material **536** may be provided at the ends of the pivoting foot sections **428** such that the flexible material **536** contacts the fibrous preform during the needling process. In various embodiments, flexible material **536** comprises a flexible material such as silicone or rubber, among others.

(43) With reference to FIG. **11**, an end effector **602** is illustrated having an array of retractable presser feet **606** disposed adjacent and alternating with an array of retractable needles **608**. In this manner, although various embodiments of the present disclosure are illustrated with needles coaxially aligned with (e.g., extending through) corresponding presser feet, it is further contemplated that the needles **608** be disposed adjacent to the presser feet **606**. The presser feet **606** may be programmed to conform and compact the surface of the fibrous preform **210**. The needles **608** may be programmed to penetrate the fibrous preform **210** to a desired depth.

(44) With reference to FIG. **12**, an end effector **702** is illustrated having an actively controlled array of retractable presser feet **706**. In this manner, actuators **722** may be provided for controlling a position of the retractable presser feet **706**.

(45) With reference to FIG. **13**, an end effector **802** is illustrated having a passively controlled array of retractable presser feet **806**. In this manner, tuned spring members **834** may be provided for biasing presser feet **806** toward extended positions.

(46) With reference to FIG. **14** and FIG. **15**, end effectors **902** and **903**, respectively are illustrated having preform tensioners **936** and **937**, respectively, in accordance with various embodiments. In various embodiments, preform tensioners **936** and **937** may be provided in place of, or in addition to, the presser feet as described herein. In various embodiments, preform tensioners **936** and **937** comprise leaf springs **938** and **939**, respectively, having rollers **940** and **941**, respectively, disposed at the ends thereof. In various embodiments, preform tensioners **936** and **937** are disposed at opposite ends of the needles **908** and **909**, respectively. Preform tensioners **936** and **937** may provide localized preform tension during the through thickness reinforcement process. In this regard, rollers **940** and **941** may be configured to contact the fibrous preform, thereby causing the leaf springs **938** and **939** to flex and exert a tension force on the fibrous preform.

(47) Systems and methods of the present disclosure include a tool for producing composite preforms with tailored in-plane and interlaminar properties. Systems and methods of the present disclosure enable the ability to needle on a complex contour preform. Systems and methods of the present disclosure allow for precisely controlling and programing needling location, angle, depth, and areal density. Systems and methods of the present disclosure allow spatially varying the needling parameters to vary interlaminar versus in-plane properties based on the desired application. Systems and methods of the present disclosure enable fabrication of 2.5D and 3D complex contour composite preforms for aerospace structures. Systems and methods of the present lend themselves to fully automated fabrication to reduce costs, improve reproducibility, and scale to production rates.

(48) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings

between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(49) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic.

Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(50) Numbers, percentages, or other values stated herein are intended to include that value, and also other values that are about or approximately equal to the stated value, as would be appreciated by one of ordinary skill in the art encompassed by various embodiments of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable industrial process, and may include values that are within 10%, within 5%, within 1%, within 0.1%, or within 0.01% of a stated value. Additionally, the terms “substantially,” “about” or “approximately” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, the term “substantially,” “about” or “approximately” may refer to an amount that is within 10% of, within 5% of, within 1% of, within 0.1% of, and within 0.01% of a stated amount or value.

(51) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is intended to invoke 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(52) Finally, it should be understood that any of the above described concepts can be used alone or in combination with any or all of the other above described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. An end effector for through thickness reinforcement of a fibrous preform, the end effector comprising: a body; a first presser foot and a second presser foot mounted to the body and moveable with respect to the body; a first needle and a second needle mounted to the body and moveable with respect to the body, wherein the first needle is configured to move between a first extended position and a first retracted position; a first sensor configured to detect whether the first presser foot has contacted the fibrous preform and whether the first presser foot is exerting a first predetermined force on the fibrous preform; and a second sensor configured to detect whether the second presser foot has contacted the fibrous preform and whether the second presser foot is exerting a second predetermined force on the fibrous preform, the second predetermined force being different from the first predetermined force, wherein the first needle extends through the first presser foot and the second needle extends through the second presser foot.
2. The end effector of claim 1, wherein the first presser foot is configured to articulate from the body between a second extended position and a second retracted position.
3. The end effector of claim 2, wherein the second presser foot is configured to articulate from the body between a third extended position and a third retracted position, wherein the first presser foot is configured to articulate independent of the second presser foot with respect to the body.
4. The end effector of claim 1, wherein the first presser foot is moveable with respect to the first needle.
5. The end effector of claim 1, wherein the first presser foot comprises a pivoting foot section pivotally mounted to a spherical joint.
6. The end effector of claim 5, further comprising a flexible material provided at an end of the pivoting foot section, wherein the flexible material extends between adjacent pivoting foot sections.
7. The end effector of claim 1, further comprising a spring member configured to bias at the first presser foot.
8. The end effector of claim 2, further comprising an actuator for moving the first presser foot between the second extended position and the second retracted position.
9. The end effector of claim 1, further comprising an actuator for moving the first needle between the first extended position and the first retracted position.
10. The end effector of claim 1, further comprising a preform tensioner configured to contact the fibrous preform to apply tension to the fibrous preform while the needle moves between the first extended position to the first retracted position.
11. The end effector of claim 1, further comprising a robotic arm and wherein the body is coupled to the robotic arm.
12. The end effector of claim 1, wherein the first sensor and the second sensor are each multi-axial sensors and each configured to measure force in at least two directions.
13. The end effector of claim 3, wherein the independent articulation allows for the third extended position to be greater than the second extended position in response to the first presser foot and the second presser foot contacting a curved preform.
14. The end effector of claim 1, wherein the first needle and a second needle are configured to extend toward and to contact the preform at a non-orthogonal angle.
15. The end effector of claim 1, wherein the first needle is configured to penetrate a first predetermined number of layers of the fibrous preform, wherein the second needle is configured to penetrate a second predetermined number of layers of fibrous preform, the second predetermined number being greater than first predetermined number.
16. A method for performing a process of through thickness reinforcement in a fibrous preform, the method comprising: moving the end effector of claim 1 with respect to the fibrous preform; moving the end effector toward the fibrous preform until the first presser foot of the end effector contacts

the fibrous preform; moving the first needle of the end effector from the first retracted position to the first extended position; penetrating the fibrous preform with the first needle in response to the first needle moving from the first retracted position to the first extended position; and moving at least a portion of a fiber from a first layer of the fibrous preform into a second layer of the fibrous preform in response to the first needle penetrating the fibrous preform.

17. The method of claim 16, further comprising: receiving, by a control unit, a feedback signal from the first sensor indicating a pressure exerted by the first presser foot onto the fibrous preform.

18. The method of claim 16, further comprising: orienting the first needle at a non-orthogonal angle with respect to the fibrous preform; and moving the second presser foot of the end effector from a second retracted position to a second extended position.

19. The method of claim 16, further comprising: receiving, by a control unit, a feedback signal from the first sensor indicating a pressure exerted by the first needle onto the fibrous preform.
